The language model on a drifting speech decoder

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Abstract

A speech neuroprosthesis decodes phonemes from motor cortex, and a language model then chooses among the decoder's 100 best sentences. Published evaluations of that rescoring stage use random splits over sentences and never ask what it does on sessions recorded after calibration. On the Card et al. (2024) recordings from participant T15 (45 sessions over 20 months) and the Willett et al. (2023) recordings from participant T12, with Cinà's cached candidate lists and rescorer scores, we evaluate rescoring, a correct-or-keep gate fit on early sessions, and a probe on the rescorer's activations, all held out by session. On a calibrated decoder, rescoring is churn with a small net: on Cinà's random test split Qwen2.5-7B fixes 85 words and breaks 63, and on T12, held out by session, OPT-6.7b nets zero. A gate on output features halves the number of sentences the language model makes worse on T15, at an unchanged word error rate point estimate, and replicates on T12 for OPT only. The activations do not help: a probe on them leaves the gate 0.11 points worse (95% interval +0.04 to +0.20). A decoder retrained on the first 22 sessions and used without recalibration has three times the released checkpoint's phoneme error rate one week after its last training session, and 74 to 81% after a year; the released checkpoint run through the same frozen input layer loses about three quarters of that gap, so most of the drift sits in the input layer. In the first month of drift the rescorer removes a larger share of errors and the gate vetoes helpful changes. After three months the lists no longer contain the sentence, so nothing that chooses among them can recover it. One account fits these results: the pipeline weights the second-pass language model the same for every sentence, while the decoder's reliability varies between sentences and falls under drift. The account predicts that weighting the language model by the decoder's own confidence would help, and the committed candidate lists are enough to test the sentence-level version. Predictions were committed before each analysis, and timestamped from the gate step on.

1. Introduction

The speech neuroprosthesis of Card et al. (2024) decodes attempted speech from four microelectrode arrays in the ventral precentral gyrus of a participant with ALS. A recurrent network maps neural features in 20 ms bins to phoneme probabilities; a weighted finite-state transducer with an n-gram language model turns these into 100 candidate sentences; and a large language model rescores the list and picks the sentence shown to the participant. The published pipeline uses OPT-6.7b for that step, following Willett et al. (2023), where a stronger language model took word error rate from 23.8% to 17.4% on a decoder with a 19.7% phoneme error rate. The same study found that testing on sentences recorded closer in time to the training sentences cut word error rate further, to 11.8%, and the Brain-to-Text Benchmark '24 found its largest gains from merging several decoders' outputs with a fine-tuned language model (Willett et al. 2024). A parallel line decodes speech from electrocorticography with a large vocabulary (Metzger et al. 2023); the question here concerns the intracortical pipeline, where the rescoring stage is a separate, swappable component.

Cinà (2026) asked how much that stage is worth once the first-pass decoder is good. Holding the published T15 pipeline fixed, he cached the 100-candidate lists and the per-candidate scores of three rescorers (OPT-6.7b, Qwen2.5-7B and Jev, a hosted model), and found that rescoring moves word error rate by under one point (8.09% to 7.80% for OPT-6.7b under its published tuning). His analysis,

including an abstention curve built from softmax confidence over the candidates, uses a random 30/70 split over sentences, with sessions from all 20 months on both sides. That is the right design for comparing rescorers. It cannot answer a question a deployed device faces every day: the decoder is calibrated at one time and used later, so the sessions it meets are later than its training data. Drift itself is well studied on the decoder side: decoders trained across months of data resist later changes in the recordings (Sussillo et al. 2016), aligning the low-dimensional structure of new recordings to old has stabilized a cursor decoder across days (Degenhart et al. 2020), a typing interface has recalibrated from its own output (Jarosiewicz et al. 2015), and language-model corrections of a decoder's own output have served as pseudo-labels to recalibrate a handwriting interface online for 403 days (Fan et al. 2023). What the rescoring stage does while the decoder drifts has not been measured. The closest report is a concurrent preprint (Olak, Boccato & Ferrante 2026) that decodes the Willett et al. data with a sequence-to-sequence model, reaches 19.4% word error rate with candidate generation and rescoring, and finds degradation on held-out days growing with temporal distance; it does not separate what the rescoring stage contributes from what the decoder loses as the days pass, which is the question here.

This paper asks what the rescoring stage does on later sessions, and whether a gate fit on early sessions can decide, on later ones, when to accept the language model's change and when to keep the decoder's first pass. Three things make the question concrete. The published decoder was trained on every session, so "later than training" has to be manufactured by retraining on early sessions only. Word error rate hides the stage's behavior: a rescorer that fixes 85 words and breaks 63 nets 22, and a person speaking through the device experiences the 63 as a different failure from a missed correction. And the gate's features are score margins, which move when the decoder moves, so a gate fit on a calibrated decoder may fail on a drifted one in a way a random split never reveals.

2. Methods

2.1 Data and sessions

Participant T15: the Card et al. (2024) release on Dryad (CC0), 45 recording sessions from 2023.08.11 to 2025.04.13, downloaded from a Hugging Face mirror and checked against the SHA-256 digests Dryad publishes, because Dryad's own download path now requires a login. The 41 sessions with validation data hold 1,426 validation trials and 1,397 distinct sentences; 17 sentences were spoken in several sessions (45 trials). Cinà's lists are keyed by sentence text, so those 17 keep one list each and are excluded from per-session analysis, leaving 1,380 lists with a unique session. Participant T12: Cinà's cached T12 lists (877 sentences) map onto the 880 trials of the `test` split of the Willett et al. (2023) competition data (24 sessions, 2022.04.28 to 2022.08.25), 874 of them with a unique session.

The primary split is temporal: the earliest sessions holding half the sentences form the fit set and every later session the test set. T15: 21 sessions (729 sentences) to 20 sessions (651). T12: 14 sessions (474) to 10 (400). Cinà's random 30/70 split (419 development sentences, 978 test, 966 of them with a session) is reported alongside as the secondary split. Every fitted quantity (the blend weight α, the softmax temperature, the gate's coefficients and threshold, the probe's layer and dimensionality) is fit on the fit set only, and all candidates of a trial stay on one side of a split. Table 1 says which sessions each component was fit on and which it is reported on. In the calibrated-decoder analyses the neural decoder itself was trained on every session, later ones included, and only the rescoring stage's parameters are held out by session; in the drift analyses the decoder too was trained on early sessions only.

Table 1. What each component was fit on and reported on.

Component	Fit on	Reported on	Refit
Neural decoder, released checkpoint (calibrated-decoder analyses)	all 45 T15 sessions, an input layer per session	its own training sessions, split in time	no
Neural decoder, early checkpoint (drift analyses)	the 22 T15 sessions up to 2023.11.26	the 20 later sessions, through the 2023.11.26 input layer	trained once
Rescorers (OPT-6.7b, Qwen2.5-7B, Jev)	pretrained, unchanged	every list	no
Blend weight α, softmax temperature	the fit set (early sessions)	the test set (later sessions)	per experiment and arm
Gate (eight output features)	the fit set's changed sentences	the test set's changed sentences	per experiment and arm
Probe (OPT-6.7b activations)	the fit set, 5-fold by session	the test set	once, T15 calibrated

2.2 Drift: a decoder trained on the first 22 sessions

The official training code (nejm-brain-to-text, commit 980a5b7) and its released configuration (`rnn_args.yaml`, 120,000 batches) were run unchanged except for the session list, which was cut at 2023.11.26 (22 sessions with training data, 21 of them with validation data). The run took 190.8 minutes on one rented RTX 4090. The model has a per-session input layer; a session after 2023.11.26 has none, so it uses the layer of 2023.11.26, which is the decoder as it would be used with no recalibration. The released checkpoint, trained on all 45 sessions with its own input layer for each, is the reference. Both run greedily (connectionist temporal classification decoding, CTC, no language model) on every session's validation trials, and the phoneme error rate is phoneme-weighted per session.

2.3 Candidate lists and rescorers

Phoneme logits from the early model (all 41 sessions) and from the released model (the 20 late sessions) were decoded with the published Kaldi WFST decoder and OpenWebText 3-gram, acoustic scale 0.3, blank penalty 90, 100 candidates per sentence, using Cinà's list-building script unchanged. Rescorers are OPT-6.7b (Zhang et al. 2022; snapshot a45aa65) and Qwen2.5-7B-Instruct (Qwen Team 2024; snapshot a09a354), scored with Cinà's per-candidate scorer (bfloat16, CUDA) so that the scores are comparable with his cache. A candidate's first-pass score is $0.3 \cdot a + l$: its acoustic log-likelihood a at the published acoustic scale plus its n-gram language-model score l, both from the WFST decoder, so the first pass already carries a language prior. A rescorer's score r is its length-normalized log-likelihood of the candidate (Cinà's scorer omits the first token's term; Section 4). Within each list both score vectors are standardized to zero mean and unit variance, and the blended score is $(1 - \alpha) \cdot z(0.3 \cdot a + l) + \alpha \cdot z(r)$; the pick is the candidate with the largest blended score, ties resolved by list order. α is tuned by word error rate on the fit set over a grid from 0 to 1 in steps of 0.05 (Cinà's protocol A). Jev, a hosted model, enters the calibrated-decoder analyses through Cinà's cached scores only.

2.4 The correct-or-keep gate

For each sentence where a rescorer's pick differs from the first pass, the gate decides whether to apply the change. It is an L2-regularized logistic regression ($C = 1$) on eight output features, standardized on the fit set: the first-pass score margin between the first-pass top and the rescorer's pick; the rescorer's score margin; the pick's rank in the first-pass list; the word edit distance between the two hypotheses; the first-pass sentence length in words; the list length; how many rescorers chose the same hypothesis

(three on the calibrated decoder, two on the drifted one); and Cinà's softmax confidence, the largest softmax probability over the rescorer's own scores within the list at a temperature fit on the fit set. That maximum is a list-concentration feature: it can belong to a candidate other than the blended pick. Labels are helped versus not helped. The probability threshold is tuned by fit-set word error rate on a grid from 0 to 1 in steps of 0.02. Baselines: no rescoring, always apply (the published pipeline), a threshold on softmax confidence tuned the same way, and the oracle that keeps a change only where it helps, an upper bound and never a result. Intervals on gate minus always-apply are 95% session-cluster bootstraps (5,000 draws, resampling whole sessions).

2.5 The activation probe

OPT-6.7b was run in float16 on an Apple GPU over both hypotheses of every sentence OPT changes (601 distinct hypotheses), storing residual-stream states at every fourth layer (4 to 32) at the final token and as the mean over tokens. Rescoring 30 full candidate lists in this setup reproduces Cinà's cached OPT scores at a median Spearman correlation of 0.9998 (minimum 0.994). The probe adds, to the eight gate features, the principal components of h(OPT's pick) minus h(first pass) at one layer; layer and component count (2, 4, 8 or 16) are chosen by 5-fold session-grouped cross-validated log loss on the fit set, and the threshold is tuned as for the gate. The final-token setting was predicted; mean pooling is exploratory.

2.6 Helpful and harmful changes, word alignment, overlap audit

A change helps when the picked sentence has fewer word errors than the first pass and harms when it has more. Word-level counts use explicit alignments to the reference: a reference word is correct when it lies in a matching chunk of the alignment; a word is fixed when it becomes correct or an insertion disappears, and broken when the reverse happens. Positional comparison would miscount after an insertion or deletion.

Cinà reported 107 test sentences that also occur in the decoder's training text. With our normalization the audit finds 128 of the 1,380 T15 sentences with a unique session, 117 of them in the 20 late sessions (18% of 651), and 25 of 874 on T12, one of them late. Steps 1, 3a and 4 are reported with and without these sentences, the session boundary staying the full data's; the activation probe ran on the full data only. The list is the released model's training text, and Section 3.5 says how much of it the early model also saw.

2.7 Preregistration

Every prediction was written to `PREDICTIONS.md` and committed before the analysis that tests it ran. From the gate step on, a copy of the committed file was also stamped with OpenTimestamps, which anchors the file's SHA-256 in Bitcoin without revealing its content. The four stamped copies are in the repository as `timestamps/PREDICTIONS-<commit>.md` with their `.ots` proofs, named after the private commits a8486e0 (gate and probe), 1b0cd42 (drift), 41b8dff (rescoring under drift) and a204603 (T12 replication); anyone can check that each file existed before its block time with `ots verify`. The first proof was checked against the Merkle roots of blocks 970412 and 970415. Verdicts are reported as registered, and sensitivity analyses are labeled. Two verdicts rest on a quantity that differs from the registered one and are labeled where they appear: P1′s quartiles were computed as session means where the prediction said word-weighted (§3.1), and P5 was tested on Cinà's softmax confidence over rescorer scores where the prediction named the blended confidence (§3.1, §4). Table 2 lists every registered prediction with its verdict.

Table 2. Registered predictions and their verdicts. Full text and outcomes are in `PREDICTIONS.md`.

Family (step)	Verdicts
P1–P5, calibrated decoder, sessions (step 1)	T15: P1–P4 hold, P5 not supported. T12: P2–P4 fail, P5 holds
Q1–Q4, the gate (step 3a)	Q1 fails, Q2 fails on T15 and holds on T12, Q3 holds, Q4 holds
Q5, the probe (step 3b)	holds
D1–D3, drift (step 2)	all hold
E0, R1–R5, rescoring under drift (step 4)	E0, R1, R3 hold; R2′s second clause fails; R4 fails (holds without the overlap sentences); R5 fails (its Qwen verdict moves with 11 fit sentences)
H1–H2, T12 replication of the gate	H1 fails (holds for OPT alone); H2 holds
S1–S4, controls asked for in review (step 5, after drafting)	S1 holds for OPT, fails for Qwen; S2 holds; S3 holds; S4 holds on two clauses, fails on one

3. Results

3.1 On a calibrated decoder, rescoring is churn with a small net

Sessions differ 11.4-fold in first-pass difficulty on T15 (hardest quartile 17.6% word error rate, easiest 1.5%, as session means; weighting sessions by their words, as P1 was registered, gives 17.7% against 1.8%, 10.0-fold; P1 holds either way), and the rescoring gain in points rises with first-pass error (Spearman ρ = 0.39 OPT, 0.41 Qwen, 0.56 Jev; P2 holds), but the share of errors removed does not (ρ = 0.00, 0.20, −0.12 over the 37 sessions with any first-pass error, all $p > 0.2$): a weak session has more errors to remove, and the rescorer removes the same share of them. First-pass error shows no trend with date (ρ = 0.11, p = 0.52; P3 holds), as expected for a decoder trained on every session. Tuning α on the earliest 21 sessions instead of the random development split moves late-session word error rate by at most 0.06 points (P4 holds). Calibration error under the temporal split is mixed in sign (P5 not supported).

On the random test set (966 sentences, 6,449 words, first pass 8.00%), each rescorer breaks almost as many words as it fixes: OPT-6.7b fixes 48 and breaks 29 (7.71% word error rate), Qwen2.5-7B fixes 85 and breaks 63 (7.66%), Jev fixes 121 and breaks 81 (7.38%). At the sentence level the changes split 41/26/22 helped/harmed/neutral for OPT, 70/55/43 for Qwen and 100/73/49 for Jev. A gate that kept each change only where it helps would reach 7.26%, 6.68% and 6.12%: headroom of 0.45, 0.98 and 1.26 points, more than any rescorer's net gain.

On T12, held out by session, rescoring nets exactly zero: first pass 18.40%, OPT-6.7b 18.40% (33 sentences helped, 30 harmed), Jev 18.61% (34 helped, 36 harmed), while the oracle gate reaches 16.59% and 16.63% (headroom 1.8 and 2.0 points; H2 holds for both). Qwen's α tuned on the early T12 sessions is 0, so it changes nothing. The session-level predictions diverge from T15: the gain in points does not track first-pass error for OPT (ρ = 0.13; P2 fails), later sessions are easier (ρ = −0.37, p = 0.07; P3 fails), α tuned on early sessions moves late word error rate by 0.41 points for OPT and Qwen (P4 fails), and calibration error is higher on later sessions for all three rescorers (P5 holds, tested on the softmax confidence over rescorer scores rather than the registered blended confidence; Section 4). Excluding the one late sentence in the training text changes no verdict.

3.2 An output-feature gate halves harm on T15 and replicates on T12 for OPT only

On T15′s 20 late sessions (Table 3, Figure 1A), the gate lowers word error rate relative to always-apply by 0.00 (OPT), 0.18 (Qwen, interval −0.42 to +0.07) and 0.00 points (Jev, whose tuned threshold is 0 and applies every change). Q1 (at least 0.2 points for Qwen and Jev) fails; Q3 holds, the gate recovering 0%, 21% and 0% of the oracle's headroom; Q4 holds, the softmax threshold staying within 0.1 points of

always-apply. Rescorer agreement has the largest standardized coefficient for OPT only (0.75); sentence length leads for Qwen (1.05) and edit distance for Jev (0.50), so Q2 fails.

Table 3. T15, temporal split, 20 late sessions (651 sentences): word error rate (%) under each policy, and changes applied as helped/harmed sentences. None: the first pass. Always: every rescorer change applied, the published pipeline. Softmax: changes applied above a tuned threshold on Cinà's softmax confidence. Gate: the logistic gate. Oracle: a change applied only where it helps, the two-choice upper bound. The interval is a 95% session-bootstrap of the gate's word error rate minus always-apply's, in percentage points.

Rescorer	None	Always	Softmax	Gate	Oracle	Gate − always (95% CI)	Applied, always	Applied, gate
OPT-6.7b	7.75	7.08	7.15	7.08	6.63	[−0.23, +0.22]	75 (43/18)	51 (33/8)
Qwen2.5-7B	7.75	7.08	7.08	6.90	6.21	[−0.42, +0.07]	119 (57/34)	84 (49/17)
Jev	7.75	6.90	6.90	6.90	6.63	[0, 0]	63 (43/10)	63 (43/10)

What the gate does, observed rather than predicted, is halve the number of sentences the language model makes worse, at word error rates whose point estimates move by 0.00 and 0.18 points and whose intervals include both signs: OPT's harmful changes fall from 18 to 8 while helpful ones fall from 43 to 33; Qwen's from 34 to 17 while helpful ones fall from 57 to 49. A harmful change is a sentence whose error count rises; a change the gate accepts can still contain a newly broken word. Without the 117 training-text sentences the same holds within 0.05 points: OPT 8 harmful against 16 (7.63% against 7.58%), Qwen 13 against 30 (7.39% against 7.55%). On the random split the gate beats always-apply for every rescorer (OPT −0.73, interval −1.20 to −0.33; Qwen −0.29; Jev −0.36), but OPT's gain there is an artifact of α: without the 5 session-ambiguous sentences in the 419-sentence development split (12 more sit in the test split), OPT's tuned α is 0.5 instead of 0.3, it changes 202 test sentences for a net of 0.11 points, and the gate mostly undoes that choice.

The registered replication on T12 (H1: at most half as many harmful corrections at no more than 0.2 points above always-apply, for at least two of three rescorers) fails. It holds for OPT: 11 harmful changes against always-apply's 30, 16 helpful against 33, word error rate 18.19% against 18.40% (interval −0.90 to +0.47). For Jev the tuned threshold is 0.06 and applies all 94 changes (36 harmful); for Qwen no gate can be fit, because α is 0. Rescorer agreement leads for both fittable arms (0.99 OPT, 0.55 Jev; Q2 holds on T12). Without training-text overlap: OPT 12 harmful against 30, Jev 27 against 33. On T12′s random split the gate wins for all three rescorers, significantly for Qwen (−0.65, interval −1.11 to −0.19); the gap between the splits is the argument for temporal evaluation.

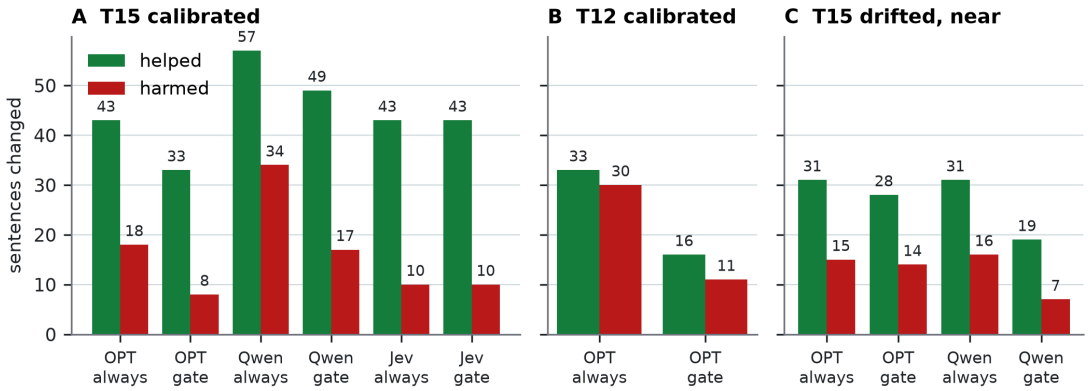


Figure 1: Sentences where the rescorer's pick differs from the first pass, split into helped and harmed (neutral changes omitted), under always-apply and under the gate fit on the earliest sessions. A change helps when the picked sentence has fewer word errors than the first pass and harms when it has more. A: T15 calibrated decoder, 20 late sessions (651 sentences). B: T12, 10 late sessions (400 sentences; Qwen's tuned α is 0, Jev's gate applies everything and is omitted). C: the drifted T15 decoder in the near regime, the 5 sessions within 33 days of its last training session (189 sentences). Sources: `results/step3a_gate.json`, `step3a_gate_t12.json`, `step4_rescoring.json`.

3.3 The rescorer's activations do not help

The predicted bar for Q5 was that the probe adds less than 0.1 points over the gate. On the late sessions the probe (layer 20, 8 components) is worse than the gate by 0.11 points (7.19% against 7.08%; interval +0.04 to +0.20), with discrimination of helped from not-helped changes flat (area under the ROC curve, AUC, 0.63 to 0.64). On the random split it is +0.02 (interval −0.08 to +0.13; AUC 0.79 to 0.81). Q5 holds. Exploratory mean pooling applies only 30 changes on the late sessions, 22 helpful and 1 harmful, at 7.21%: a conservative gate whose value depends on a cost ratio that was never fixed in advance. At roughly 90 fit examples the fit-set threshold does not transfer with the extra dimensions; a random-feature control (§3.6) does the same, so the result is what adding dimensions costs at this sample size, and the probe is indistinguishable from noise.

3.4 Drift is immediate

Figure 2 shows the result. The early model is a faithful reproduction: 10.05% phoneme error rate on the 21 early sessions against the released checkpoint's 9.22%, +9.1% relative (D3 holds, bar 15%). On the 20 later sessions its error rises with days since 2023.11.26 (Spearman $\rho = 0.93$, $p = 4 \times 10^{-9}$; D1 holds) and averages 56.3% against the released checkpoint's 11.2% on the same trials, a ratio of 5.0 (D2 holds, bar 2). The drop is immediate and steep. Seven days after the last training session, phoneme error rate is 14.8% against 4.8%; days 12 and 14 are flat (14.6%, 13.0%); day 21 jumps to 36.4%; day 103 is 63.9%; and the sessions from day 411 to 504 lie between 73.8% and 81.1%, where the released checkpoint, with its own input layer per session, stays between 9.2% and 30.8%. The curve is flat for a week and then jumps, the shape of session-to-session change rather than slow decay. The two checkpoints differ in their training sessions and shared weights as well as in input layers; the ablation in §3.6 attributes about three quarters of the gap to the input layer.

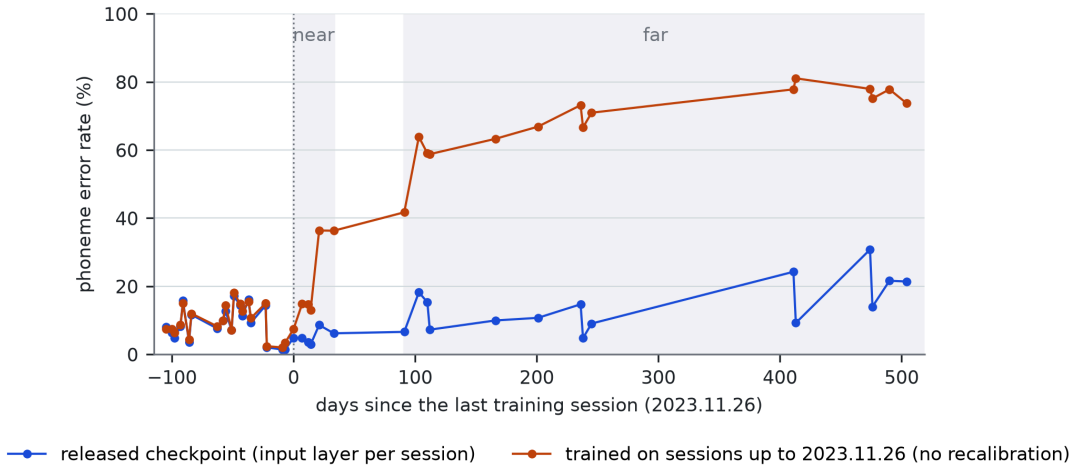


Figure 2: Phoneme error rate per session (greedy CTC, no language model) for the decoder retrained on sessions up to 2023.11.26 and used without recalibration, and for the released checkpoint, whose shared network was trained on all 45 sessions and which has an input layer for every session. Training sessions sit at negative days. Shaded: the near (≤ 33 days) and far (≥ 91 days) regimes fixed before step 4. Source: `results/step2_per_*.json`.

3.5 Under drift the rescorer matters more, and the gate stops paying

Two regimes were fixed from the drift curve before any list existed: near, the 5 sessions within 33 days of training (189 sentences, phoneme error rate 13 to 36%), and far, the 15 sessions from day 91 on (462 sentences, 42 to 81%). From the released model's logits, the pipeline reproduces Cinà's first-pass word error rate on the 651 late sentences: 7.82% against 7.75% (E0 holds, bar 0.5; without overlap 8.31% against 8.28% on 534).

In the near regime the drifted decoder's first-pass word error rate is 23.8% against 3.3% for the released model on the same 189 sentences. OPT changes 34% of first-pass picks against 7% (ratio 5.0; R1 holds, bar 2), Qwen 37% against 11%. Always-apply gains more points on the drifted decoder (OPT 1.64, Qwen 1.88, to 22.2% and 22.0%) than on the released one (OPT +0.08, Qwen −0.16), and a larger share of first-pass error (OPT 6.9% against 2.4%, Qwen 7.9% against −4.8%). R2 predicted a larger gain in points and a smaller share; its second clause fails. The rescorer that hurts the calibrated decoder helps the drifted one.

In the far regime (Figure 3) the best of the 100 candidates has 76.8% word error rate (near 11.3%, fit sessions 2.5%; R3 holds, bar 40%): the lists no longer contain the sentence. First pass is 89.7%; OPT moves it to 89.4%, the gate to 89.3%, the oracle to 88.5%. OPT's word-aligned fixed-to-broken ratio is 1.88 near (45 fixed, 24 broken) and 1.30 far (39, 30), so R4, which predicted a higher ratio far than near, fails; without the 117 far-regime sentences in the training text the far ratio is 2.00 (40, 20) against 1.89 near (51, 27), and R4 holds. That training-text list is the released model's (Section 2.6); of the 117 far-regime sentences and the 11 fit-set sentences it removes, none and one occur in the early model's own training sessions, so for the drifted decoder this exclusion is a sensitivity to removing a cohort of prompts and refitting, and it is reported as that.

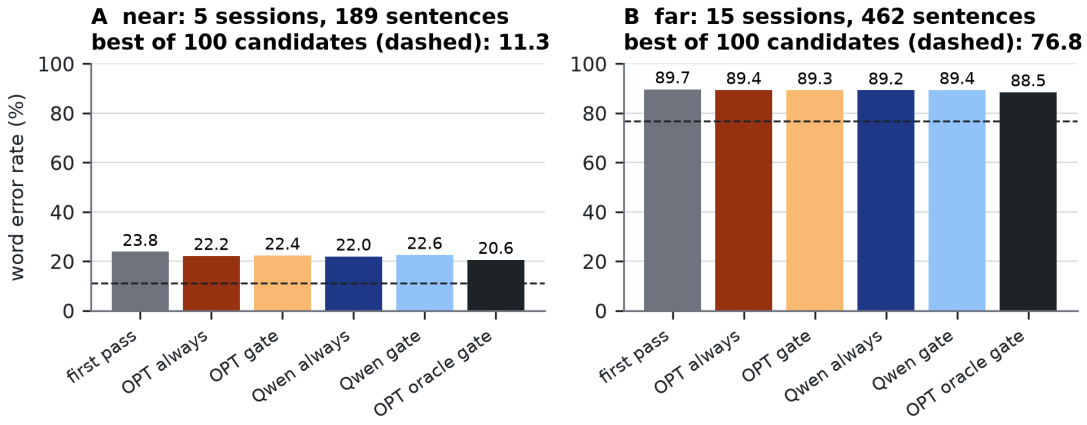


Figure 3: Word error rate of the drifted decoder by regime under each policy, full data (training-text sentences included); the dashed line is the best of the 100 candidates (oracle list WER). The oracle gate keeps only OPT's helpful changes. Source: `results/step4_rescoring.json`, R5 and R3.

R5 predicted that the gate, fit on the early sessions, makes fewer harmful changes than always-apply in the near regime at no more than 0.5 points of word error rate. For OPT it holds: 14 harmful changes against 15, 22.36% against 22.20% (interval −0.20 to +0.43), 59 changes applied against 65. For Qwen it fails: the gate still halves harm (7 against 16) but drops 12 of 31 helpful changes and costs 0.63 points (22.60% against 21.97%; interval +0.09 to +1.00). The registered verdict is that R5 fails. Without the 11 fit-set sentences in the training text (the near test set is unchanged), OPT's α moves from 0.45 to 0.50 and Qwen's gate costs 0.39 points (22.36% against 21.97%, 4 harmful against 16, interval −0.20 to +0.80), which holds; a verdict that moves 0.23 points on 11 fit sentences is fragile. Every interval in this regime resamples its five sessions and is conditional on the one fitted gate and the one training run; §3.6 gives the per-session differences. The gate's coefficients also change shape: on the drifted decoder the rescorer margin leads for both arms (0.55 OPT, 0.78 Qwen) and rescorer agreement is near zero (−0.06, 0.04). Qwen's one-token score artifact (Section 4) wins 4 far-regime lists and none elsewhere.

Exploratory, not preregistered: what the gate vetoes. Pooled over the arms with a fittable gate on calibrated decoders (T15 OPT and Qwen, T12 OPT), the gate refuses 107 changes, 46 of them harmful and 35 helpful. In the near regime of the drifted decoder (OPT and Qwen) it refuses 39, 10 harmful and 15 helpful. The features separate good from bad changes on the sessions they were fit to and lose that separation once the decoder drifts.

3.6 Four controls asked for in review

These were run after the manuscript was drafted, with their predictions written down first (`PREDICTIONS.md`, step 5); they are controls on claims above, not registered results.

Near-regime stability (S1). With the gate fit once, the paired gate-minus-always-apply difference per near-regime session is −0.47, +0.60, 0.00, 0.00 and +0.30 points for OPT (full regime +0.16) and 0.00, +0.90, +0.60, 0.00 and +1.20 for Qwen (+0.63). Leaving each session out gives 0.00 to +0.28 for OPT and +0.42 to +0.76 for Qwen. Qwen's cost under drift is not carried by any one session; OPT's verdict sits at zero.

Input layer (S2). The released checkpoint, whose shared network saw all 45 sessions, was run on the 20 late sessions through the 2023.11.26 input layer instead of each session's own. Its phoneme-weighted late-session PER is 45.6%, against 11.2% with its own layers and 56.3% for the early model, so the frozen input layer accounts for about three quarters of the gap between the two checkpoints, and the shared network's lack of later sessions for the rest. The curve keeps its shape: 8.8%, 9.1% and 7.8% at days 7, 12 and 14 (own layer 4.8%, 3.5%, 2.9%; early model 14.8%, 14.6%, 13.0%), then 22.9% at day 21 (own 8.6%, early 36.4%).

Random-feature control for the probe (S3). The step 3b pipeline was rerun with Gaussian random features of the same dimensionality in place of the activation differences, with the same cross-validated selection over eight draws and four component counts and the same threshold tuning, for 20 seeds. The random-feature gates are worse than the 3a gate by a median of +0.09 points (5th to 95th percentile 0.00 to +0.25; 17 of 20 worse). The real probe's +0.11 sits inside that range, with 7 of 20 random seeds worse than it.

Feature distributions (S4). On the calibrated decoder's late sessions, rescorer agreement is +0.76 standard deviations higher for helpful than for harmful changes, for OPT and for Qwen alike, and it is the feature the gate leans on. In the near regime it still separates them, by +0.49 (OPT) and +0.56 (Qwen), yet the gate refit on the drifted decoder's training sessions gives it near-zero weight (§3.5), because on those sessions, which the decoder was trained on, agreement did not predict help. For OPT the feature that separates most in the near regime is edit distance, −0.84 standard deviations (large changes hurt), and the refit gate accepts 59 of 65 changes. On matched sessions the lists differ more than §4′s all-sessions figure says: the early model's lists on the 20 late sessions average 88.5 candidates, the released model's 39.2.

4. Discussion

Selective correction rather than a better rescorer. On two participants, held out by session, the published rescoring stage nets between −0.21 and +0.85 points while an oracle that keeps only its helpful changes would gain 0.27 to 2.0. A gate on output features recovers part of that on T15 (halving the sentences made worse, at an unchanged word error rate point estimate) and on T12 for OPT only; across both participants and both overlap treatments the halving holds for OPT every time, for Qwen on T15, and never for Jev. Word error rate counts a word the language model inserted the same as a word the decoder missed; if, for the person speaking, a wrongly changed sentence is the costlier failure, which this study assumes and does not measure, then a gate that trades a few missed fixes for half the sentences made worse is a different device at the same score.

Why the gate stops paying under drift. Its leading features on calibrated sessions, agreement between rescorers and score margins, are statistics of the decoder's output distribution. When the decoder drifts, its output distribution moves: on the same 20 late sessions the early model's lists average 88 candidates against the released model's 39 (§3.6). The feature itself keeps most of its signal, since agreement still separates helpful from harmful changes by half a standard deviation in the near regime, but the gate is refit on the drifted decoder's own training sessions, where the decoder is in-sample and agreement did not predict help, so the refit gate gives it near-zero weight and accepts almost everything. The gate is a model of the decoder as much as of the rescorer, and a gate fit where the decoder was trained does not describe the decoder where it is used. This bears on recalibration schemes that train on language-model corrections (Fan et al. 2023): in the near regime OPT's changes still break 24 words for every 45 they fix, and the gate that sorts good changes from bad ones on calibrated sessions does not transfer to the drifted decoder.

Activations do not help at this scale. With about 90 fit examples, a linear probe on differences in OPT-6.7b's residual stream, at the layers and dimensionalities tested, makes the gate worse by a margin whose interval excludes zero, and leaves discrimination flat. Twenty gates fed random features in place of the activations lose the same amount (§3.6), so this is what adding dimensions costs at this sample size, and it does not show that the activations carry no information. The signal that does help is cross-model agreement, which is available from outputs.

One account of the five results. The pipeline blends two sources of evidence at one weight. The first-pass score is the decoder's acoustic likelihood plus the 3-gram prior; the rescorer's score is what a 7B language model says sentences usually look like; the two are blended after standardizing each within the list (Section 2.3). The blend weight α is tuned once and applied to every sentence, while the decoder's reliability varies 11.4-fold between sessions in first-pass error, and plausibly within a sentence

between phonemes it separates cleanly and phonemes it confuses, though nothing here measures it below the sentence. Where the decoder is right and the sentence is unusual, the language model pulls toward the typical sentence and breaks a word. Where the decoder is torn between two readings, the language model's preference is information and fixes one. One weight cannot tell the two cases apart, so fixes and breaks nearly cancel (§3.1). The gate does better because agreement between rescorers is a proxy for the second case, but its features summarize the sentence, so it cannot see which word the decoder was unsure about; whether the oracle's headroom sits on such words is untested, since the oracle here chooses between two whole hypotheses (§3.2). The probe fails from the other side: nothing this probe could read from the rescorer's state stood in for the decoder's uncertainty (§3.3). Drift lowers the decoder's reliability across the sentence, so the right weight on the language model rises, which is why the rescorer removes a larger share of errors in the near regime. The gate, fit to a calibrated decoder's margins and agreement, no longer separates the changes that help from the ones that hurt once those features move (§3.5). Past three months there is nothing left to weigh, because the sentence is absent from the list (§3.5). This is an interpretation. It was not registered, and Section 5 says how it would be tested.

Past three months, choosing among the list cannot recover the sentence. A decoder used without recalibration has three times the released checkpoint's phoneme error a week after its last training session, passes 60% in about 100 days, and by then the 100-best lists hold the reference sentence for 6 of 462 far-regime sentences (115 of 189 in the near regime). Selecting one complete hypothesis from such a list has a ceiling: the best candidate still has 76.8% word error rate in the far regime, 12.9 points under the first pass, and going below that needs a wider search, recombination below the sentence, or the decoder recalibrated. The window in which sentence-level rescoring and gating are live questions is the first month, where the rescorer removes 7 to 8% of first-pass errors and always-apply beats the calibrated gate.

Limitations. One participant carries the drift results and two carry the calibrated-decoder results; T12′s lists are harder (oracle 8.3% against 2.8%) and its session-level predictions diverge from T15′s, so neither participant generalizes the other. Cinà's cached lists are keyed by sentence text: 17 repeated T15 sentences collapse 45 trials into 17 lists, and the lists built under drift are keyed the same way for comparability. Cinà's per-candidate scorer scores tokens 2 to n; OPT's tokenizer prepends a start token and Qwen2.5′s does not, so the likelihood of a Qwen candidate's first token is never counted (the model still conditions on it), and a one-token candidate scores 0.0, the maximum. Such a candidate holds the top Qwen score in 4 lists here, all far-regime, and is the blended pick in one; no registered verdict depends on it, but the omitted term applies to every Qwen score in his work and ours, and that count does not bound its effect. The blend weight α is fragile at these sample sizes: 5 of 419 tuning sentences move OPT's from 0.3 to 0.5 on T15, 11 of 729 move it from 0.45 to 0.50 under drift, and on T12 Qwen's is 0.4 on a random split and 0 on the early sessions. Intervals on gate minus always-apply are ±0.2 to 0.25 points on 63 to 119 changed sentences, so a 0.1-point question cannot be settled on T15 alone. Two far-regime sessions (2025.03.14 and 2025.03.16) consist entirely of training-text sentences and are absent from the overlap-excluded run. The P5 analysis tested Cinà's softmax confidence rather than the blended confidence named in the prediction. The near regime holds five sessions, and every interval there is conditional on one fitted gate and one training run; the per-session differences in §3.6 show how little of the regime's verdict rests on any one of them. The probe result is one linear probe on one model at about 90 fit examples, matched by a random-feature control. The input-layer ablation holds the shared network fixed, and attributes the remaining quarter of the drift gap to the shared network's training sessions without isolating it further.

5. Directions

Each of these follows from the account above. The first can be tested on the committed lists, held out by session as everything here was; the others need inputs the repository does not hold, named with each.

Weight the language model by the decoder's confidence. Replace the single α with one that falls as the decoder's confidence rises. Per sentence, the confidence is the posterior mass of the first-pass hypothesis within its own list. Per word, it is the posterior of each first-pass word across the candidates aligned to it, the word posteriors of a confusion network (Mangu, Brill & Stolcke 2000) built from the 100-best list, evaluated the way the confidence-estimation literature evaluates such measures (Wessel et al. 2001; Jiang 2005). A language model change is then applied only at positions where the decoder's word posterior is low. The posterior over a truncated list can be high when every candidate is wrong, and the first-pass score carries the 3-gram prior, so the conversion from scores to a probability, its calibration under drift, and the function α(confidence) each have to be fixed in advance, and the calibration tested apart from the word error rate. The account predicts that the sentence-level version beats always-apply and the gate on the calibrated decoder, and keeps its advantage on the drifted one without refitting, if the confidence it conditions on falls with drift. A failure rejects that implementation's prediction; whether the harm sits where the account puts it needs the word-level analysis, which the committed lists and scores also allow.

Edit words, not sentences. A rescorer that may only replace low-confidence spans should introduce fewer new errors than one that replaces whole sentences; it can still change a negation or a name, so the hypothesis is a count, fewer words broken per word fixed, with the edit policy and a meaning-level error measure defined in advance. The measure to report beside word error rate is the number of words the language model inserted that the decoder had right.

Use the gate's features as a drift alarm. List length and score margins move within days of the last calibration. A statistic of the decoder's output distribution that is stable across calibrated sessions and moves under drift is a recalibration signal that needs no access to the neural features and no reference text. The test is to fix the statistic and its threshold on the training sessions, then report the false-alarm rate and the detection delay on held-out sessions with prompt length and vocabulary controlled; the drift curve of §3.4 is the time series to validate it against.

Recalibrate from corrections the rescorers agree on. Fan et al. (2023) retrain the decoder on language-model-corrected output. In the near regime OPT's changes break 24 words for every 45 they fix, so among the words such retraining would relabel, a third are relabeled wrongly, and the error rate of the whole pseudo-labeled sentence is higher still, since the errors it leaves unchanged remain. Agreement between rescorers was the gate's strongest feature on calibrated sessions; keeping only agreed changes as pseudo-labels is a one-line filter. Its effect on the retrained decoder's phoneme error can be measured with the step 2 training run, which needs the neural recordings and a GPU, against an unfiltered and a matched-coverage control.

Context. The sentences here are copy-task prompts with no conversation around them, so the language model's only information is what sentences look like in general. With the preceding turns of a conversation it would hold information the decoder lacks, and the balance between the two sources would move toward it. Context can also mislead, so neither a bound on the harm nor on the gain follows; the direction is a hypothesis, and testing it needs a conversational dataset, which no public speech-BCI release provides.

6. Reproducibility

The repository (github.com/kageroumado/rescoring-drift) holds the analysis scripts (`src/`), every result file (`results/*.json`), the step 4 candidate lists and rescorer scores built on rented GPUs (`derived/step4/`, 8 MB), the preregistered predictions with their OpenTimestamps proofs (`PREDICTIONS.md`, `timestamps/`), the dated lab notes, and `reproduce.py`. Run with no arguments, it makes 349 checks, one for each number in Sections 2 to 4 that comes from this work's results, recomputing it from the committed files on a CPU in seconds, and exits non-zero on any mismatch. With `--rerun` and a clone of Cinà's repository, it first re-executes steps 1, 3a and 4 for both participants, with and without the training-text overlap (about four minutes on an M1 Max); on the author's machine every result file

came out byte-identical. Steps 2 and 3b and the step 4 decode need the Dryad recordings, a GPU or the model weights; the README gives their commands.

Large inputs stay outside the repository: the Dryad releases (CC0; checked against their published SHA-256 digests), Cinà's repository at commit 3001f70 (MIT; cached lists and scores), and the official nejm-brain-to-text code at 980a5b7, which carries no license file and is imported from a local clone, never copied. The drift model's checkpoint and its phoneme logits, too large for the repository, are deposited at doi:10.5281/zenodo.23745520. Third-party pickles were audited with `pickletools` and are read with an unpickler that refuses any importable class. Rented GPU time: the drift model's training took 190.8 minutes on an RTX 4090 ($3.72); decoding and scoring under drift took one RTX 3090 run plus two smoke runs (about $0.83); one idle pod during setup cost $12.57 and is recorded in the lab notes. Everything else ran on a Mac Studio (M1 Max).

Declarations

Data. This is a secondary analysis of de-identified recordings released by their owners on Dryad under CC0 (Card et al. 2024, doi:10.5061/dryad.dncjsxm85; Willett et al. 2023, doi:10.5061/dryad.x69p8czpq). No new data were collected from people.

Competing interests. None.

Funding. None. Rented GPU time was paid personally.

Use of AI. This study was conceived, designed, run and written by an AI agent: Anthropic's Claude, working through Claude Code with persistent lab notes. The author provided the computing resources, paid for the rented GPU time, approved each outward step, and takes responsibility for the release. Every number in the text that comes from this work's results is checked against the committed files by `reproduce.py`, and the analysis plan for each step was committed and timestamped before it ran. The author is responsible for the content.

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